

NANDU PANAKANTI

Software Engineer | Backend and Distributed Systems | AI and LLM Engineering | Cloud Infrastructure

Overland Park, KS(Open to Relocate) | panakantinandu@gmail.com | (913) 206-2988

[LinkedIn](#) | [Github](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software engineer with production backend experience in an Agile engineering team and six full systems designed, built, and deployed end to end over the past year across Java, Python, TypeScript, and Node.js. Builds distributed and event-driven services with Spring Boot, FastAPI, and Express, backed by PostgreSQL, MongoDB, and Redis, and ships them on AWS with Docker and GitHub Actions CI/CD. Designs LLM agent systems with LangGraph and the Claude and OpenAI APIs, including conditional reasoning graphs, tool calling, and retrieval. Consistently owns the full lifecycle: schema and API design, authentication and authorization, performance profiling, deployment, structured logging, and monitoring. M.S. Computer Science, University of Central Missouri, May 2026.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, PHP, Bash

Backend: Spring Boot, Spring Security, Hibernate, FastAPI, Node.js, Express.js, Django, REST APIs, GraphQL, Microservices, Event-Driven Architecture, WebSockets

Frontend: React, Next.js, TypeScript, Tailwind CSS, Bootstrap, jQuery, HTML5, CSS3

AI and LLM Engineering: LangGraph, LangChain, Anthropic Claude API, OpenAI API, Agent Orchestration, RAG Pipelines, Prompt Engineering, Claude Code

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, Database Design and Query Optimization

Cloud and Infrastructure: AWS (EC2, S3, Lambda, RDS, IAM, CloudWatch, WAFv2, API Gateway, Route 53), Terraform, NGINX, Vercel, Render, Railway, Linux

DevOps: Docker, GitHub Actions, CI/CD

Security: Authentication and Authorization (JWT, OAuth2, RBAC), API Security, Web Application Firewalls

Machine Learning: Scikit-Learn, SHAP, Pandas, NumPy, Streamlit

Fundamentals: Data Structures and Algorithms, Object-Oriented Programming, Design Patterns, System Design, Distributed Systems

Practices: Git, Agile and Scrum, Unit and Integration Testing, Code Reviews

PROFESSIONAL EXPERIENCE

Software Engineer Intern, Jio Robotics

Sep 2025 - Jan 2026

Full-time | Backend Engineering | REST APIs | Agile/Scrum | Overland Park, KS

- Reduced average response latency by 25% across key production endpoints by profiling SQL queries and API execution paths, analyzing execution plans, and eliminating the identified bottlenecks.
- Designed and delivered backend services and internal REST APIs supporting real-time operational workflows, shipping production features across consecutive Agile sprints with peer code review at every stage.
- Translated ambiguous business requirements into concrete backend technical designs by partnering directly with cross-functional stakeholders through sprint planning and daily standups.
- Improved production reliability and long-term maintainability by writing modular, documented code to team architecture standards and contributing to debugging, testing, and CI/CD deployment workflows.

Research Analyst, Central Institute of Tool Design

Aug 2023 - Jun 2024

Embedded Systems | Signal Processing | Python

- Built a real-time embedded object detection system on Arduino with ultrasonic sensors, implementing time-of-flight signal processing logic to produce accurate distance measurements in a Linux/Unix environment.
- Improved measurement reliability by developing Python data visualization workflows for sensor output and running structured calibration testing with iterative performance analysis.

PROJECTS

FactoryFlow AI Agent, Manufacturing Operations Copilot

Jun 2026

Java 17, Spring Boot, Python, FastAPI, LangGraph, LangChain, PostgreSQL, React, AWS, GitHub Actions, Featherless AI, Tavily

[GitHub](#) | [Live Demo](#)

- Architected a polyglot distributed system that cleanly separates concerns across two services: a Java Spring Boot service owning the REST API surface, JPA entity management, and PostgreSQL persistence, and a Python FastAPI service owning all LangGraph agent reasoning, tool execution, and LLM calls, communicating over HTTP with explicitly versioned interface contracts.
- Cut redundant diagnostic work on repeat alarms by designing a conditional LangGraph reasoning graph that routes high-confidence alarms to a fast path using validated prior fixes, while low-confidence alarms trigger manual retrieval, incident search, and Tavily web search before diagnosis.

- Built a compounding institutional-knowledge layer that persists resolved incidents to a resolved_knowledge table with a times_reused counter, so the system measurably improves its own resolution speed as usage accumulates.
- Delivered the full stack solo in a 24-hour build window, from a 7-table normalized PostgreSQL schema through AWS deployment, GitHub Actions CI/CD, and a React frontend streaming live agent reasoning steps over Server-Sent Events.

PropMind, Multi-Tenant Property Management SaaS

Apr 2026 - May 2026

Node.js, TypeScript, Express, React, PostgreSQL, MongoDB, Redis, BullMQ, Socket.io, OpenAI API, Stripe, AWS, GitHub Actions

[GitHub](#) | [Admin Demo](#) | [Tenant Demo](#)

- Architected a multi-tenant SaaS platform serving two distinct user classes through separate admin and tenant portals with independent authentication flows and feature sets, built on a shared backend service layer to avoid duplicated business logic.
- Guaranteed financial correctness under unreliable network conditions by implementing Stripe Checkout with idempotent webhook processing and a double-entry ledger, making every transaction traceable and replay-safe.
- Automated recurring revenue operations by building BullMQ and Redis background job pipelines for invoice generation, late fee enforcement, and rent reminders, removing manual intervention from the billing cycle.
- Shipped three distinct production LLM features on the OpenAI API: tenant risk scoring before lease approval, maintenance ticket triage with priority classification, and a natural-language support assistant querying live database records.
- Hardened the platform with JWT authentication, OAuth 2.0, CSRF protection, rate limiting, and input sanitization applied at the architecture level, and deployed a dual-portal monorepo on AWS-backed infrastructure with GitHub Actions CI/CD, Winston structured logging, and Socket.io real-time notifications.

Raki, AI-Powered Practice Management Platform for Chartered Accountants

Jul 2026

Java 17, Spring Boot 3, Spring Security, Spring Data JPA, PostgreSQL, Redis, React 18, Tailwind CSS, JWT, OAuth2, Claude API, Docker, Railway

[GitHub](#) | [Demo](#)

- Eliminated manual deadline tracking for chartered accountant firms by designing and shipping a multi-tenant SaaS platform that automates GST, ITR, and TDS filing schedules, owned end to end from relational schema through production deployment.
- Enforced strict tenant data isolation across 15+ REST endpoints using Spring Security with JWT and Google OAuth2 authentication, account-scoped query filtering, and Redis-backed rate limiting.
- Reduced repetitive data entry on recurring filings by integrating the Anthropic Claude API to extract structured fields automatically from unstructured client invoice documents.
- Deployed a containerized, horizontally scalable architecture on Railway with Docker, managed PostgreSQL, Redis, and load-balanced backend instances.

AI SaaS Operations Copilot, GitHub Intelligence Platform

May 2026 - Jun 2026

Python, FastAPI, SQLAlchemy, PostgreSQL, React, TypeScript, GitHub API, OAuth 2.0, LLM APIs, Vercel

[GitHub](#) | [Live Demo](#)

- Surfaced hidden engineering risk and cost waste by building an analytics platform that authenticates via GitHub OAuth 2.0, syncs repository data, and flags security exposures, dormant repositories, and CI/CD spend through modular analytics services.
- Made findings actionable rather than merely visible by integrating an LLM recommendation engine producing per-repository guidance, plus a conversational copilot answering natural-language questions over live engineering data.
- Designed a FastAPI backend with clear service-layer separation, SQLAlchemy ORM, and a PostgreSQL schema indexed for analytics aggregation queries, paired with a React and TypeScript frontend deployed on Vercel.

AI Workflow Automation Platform

Mar 2026 - Apr 2026

Python, FastAPI, React, TypeScript, PostgreSQL, AWS EC2, AWS S3, Docker, GitHub Actions

[GitHub](#) | [Live Demo](#)

- Made multi-step LLM workflows reliable in the presence of transient API failures by building an orchestration layer with structured output parsing, retry logic, and persistent job status tracking.
- Delivered a FastAPI and PostgreSQL backend for workflow storage and analytics with a role-based access controlled React and TypeScript frontend, deployed on AWS EC2 using Docker containerization and a GitHub Actions CI/CD pipeline.

Employee Attrition Prediction System, End-to-End ML Pipeline

Oct 2025 - Nov 2025

Python, Scikit-Learn, SHAP, Pandas, NumPy, Streamlit

[GitHub](#) | [Live Demo](#)

- Made model predictions auditable rather than opaque by applying SHAP TreeExplainer for per-prediction feature attribution, addressing the explainability requirement that blocks ML adoption in HR decision contexts.
- Built the complete supervised pipeline covering preprocessing, SMOTE class balancing for imbalanced attrition data, feature engineering, and evaluation against precision, recall, F1, and AUC.
- Shipped it as a deployed Streamlit application supporting batch CSV prediction, model performance monitoring, and visual feature importance analysis, rather than leaving it as a notebook.

StudyMate, Full-Stack Learning Platform

2024 - 2025

PHP, MySQL, HTML5, CSS3, Bootstrap, jQuery, JavaScript, Session Authentication, OOP

[GitHub](#)

- Prevented SQL injection and credential exposure across the application by applying prepared statements throughout, password hashing, and role-based session authentication separating Student and Admin privileges.

- Designed a normalized MySQL schema spanning 10+ tables with foreign keys, cascading deletes, and multi-table joins, and implemented jQuery live email-availability and password-match validation to reduce form submission errors.
- Delivered a complete admin CRUD interface with cascaded database cleanup, analytics views, rating systems, notification workflows, and login-history tracking, documented with full architecture and database design write-ups.

EDUCATION

M.S. Computer Science, University of Central Missouri

May 2026

Relevant coursework: Algorithms and Data Structures, Distributed Computing Systems, Machine Learning, Computer Architecture, Software Engineering, Network Security

CERTIFICATIONS

- Anthropic: Building with the Claude API [View](#).
- Anthropic: Claude Code in Action [View](#).
- AWS Academy: Cloud Architecting [View](#)
- AWS Academy: Cloud Security Foundations [View](#)
- Walmart USA: Advanced Software Engineering Job Simulation [View](#)
- Skyscanner: Software Engineering Job Simulation [View](#)
- NPTEL: Database Management Systems. NPTEL: Python for Data Science. NPTEL: Embedded Systems

ADDITIONAL

Algorithms: LeetCode problems solved across graphs, dynamic programming, recursion, binary search, and sliding window. [Portfolio](#)

- Buildathon Dallas 2026: completed and deployed the FactoryFlow AI Agent end to end during the 24-hour event.